

| Question  | Answer                                                                                                                                                                                                                                               | Marks     |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| 1(a)      | rate of change of velocity                                                                                                                                                                                                                           | <b>B1</b> |
| 1(b)(i)   | curved path from aircraft to ground, starting horizontal at aircraft and then with increasing negative gradient as it moves towards the ground                                                                                                       | <b>B1</b> |
| 1(b)(ii)  | $s = ut + \frac{1}{2}at^2$<br>$63 = \frac{1}{2} \times 9.81 \times t^2$                                                                                                                                                                              | <b>C1</b> |
|           | time = 3.6 s                                                                                                                                                                                                                                         | <b>A1</b> |
| 1(b)(iii) | $v^2 = 2 \times 9.81 \times 63$<br><b>or</b><br>$v = 0 + (9.81 \times 3.6)$<br><b>or</b><br>$63 = (v \times 3.6) - (\frac{1}{2} \times 9.81 \times 3.6^2)$<br><b>or</b><br>$63 = \frac{1}{2} \times (0 + v) \times 3.6$<br>$v = 35 \text{ m s}^{-1}$ | <b>A1</b> |
| 1(b)(iv)  | speed <sup>2</sup> = 35 <sup>2</sup> + 42 <sup>2</sup>                                                                                                                                                                                               | <b>C1</b> |
|           | speed = 55 m s <sup>-1</sup>                                                                                                                                                                                                                         | <b>A1</b> |

9702/23

Cambridge International AS & A Level – Mark Scheme

October/November 2024

**PUBLISHED**

| Question | Answer | Marks |
|----------|--------|-------|
|          |        |       |